

Linux Practical Assignment: TechNova Server Setup

Scenario: You are a junior system administrator at *TechNova*. Your manager asks you to set up a workspace for the new development team. Complete all tasks in a Linux terminal.

Topics covered: command prompt, file system navigation, file management, Vim, users and groups, permissions, links, sudo, tar and compression, cron.

PART 1: TASKS

Task 1: System Information

1. Display your username, hostname, and current directory.
2. Show system information, disk usage, and memory usage in MB.
3. Save all this output into a file called `sysinfo.txt`.

Task 2: Directory Structure

1. In your home directory, create the structure below using a **single** `mkdir -p` command.
2. Create empty files `app.txt` and `notes.txt` inside `projects/`.
3. Display the structure using `ls -R` (or `tree`).

```
technova/  
|-- projects/  
|-- docs/  
|-- backups/  
`-- scripts/
```

Task 3: Copy, Move, Delete

1. Copy `app.txt` to the `docs/` folder.
2. Rename `notes.txt` to `notes_old.txt`.
3. Move `notes_old.txt` to `backups/`.
4. Delete `app.txt` from `projects/` using confirmation mode.

Task 4: Vim Editor

1. Create `docs/readme.txt` using Vim.
2. Write 3 lines of text about TechNova.
3. Turn on line numbers, search for the word "TechNova", copy line 1 and paste it at the end, then undo the paste.
4. Save and quit.

Task 5: Users and Groups

1. Create a group `dev_team`.
2. Create users `alice` and `bob` with home directories and the bash shell.
3. Set passwords for both.
4. Add both to `dev_team` and verify.
5. Lock `bob`'s account and check password aging details for `alice`.

Task 6: File Permissions

1. Create `scripts/backup.sh` and make it executable by the owner only (no access for group or others).
2. Create `docs/secret.txt` so only you can read and write it.
3. Create a directory `/shared` where `dev_team` members can read, write, and enter.
4. Verify all permissions with `ls -l`.

Task 7: Links

1. Create a file `original.txt` with some text.
2. Create a hard link `hard.txt` and a soft link `soft.txt`.
3. Check the link counts and inode numbers.
4. Delete `original.txt` and observe which link still works.

Task 8: sudo and Privileges

1. Try `ls /root` as a normal user, then with `sudo`.
2. Give `alice` `sudo` access.

Task 9: Archiving and Compression

1. Create a `.tar.gz` archive of the `technova` folder contents in `backups/`.
2. List the archive contents without extracting.
3. Extract it into `/tmp/restore`.

Task 10: Automation with Cron

1. Create a cron job that makes a timestamped file in `~/technova/backups/` every minute.
2. Create a cron job that archives `~/technova/docs` every day at 2:00 AM.
3. List your cron jobs, wait 2 minutes, and verify files are being created.
4. Remove the every-minute job after testing.

Task 11: Cleanup

1. Remove users `alice` and `bob` with their home directories, and delete the group `dev_team`.

PART 2: HOW TO DO EACH TASK (Step by Step)

Task 1: System Information

```
whoami
hostname
pwd
uname -a
df -h
free -m
```

Save everything into one file:

```
{ whoami; hostname; pwd; uname -a; df -h; free -m; } > sysinfo.txt
cat sysinfo.txt
```

Task 2: Directory Structure

```
cd ~
mkdir -p technova/{projects,docs,backups,scripts}
touch technova/projects/app.txt technova/projects/notes.txt
ls -R technova
```

Task 3: Copy, Move, Delete

```
cd ~/technova
cp projects/app.txt docs/
mv projects/notes.txt projects/notes_old.txt
mv projects/notes_old.txt backups/
rm -i projects/app.txt          # type y to confirm
ls -R
```

Task 4: Vim

```
vim docs/readme.txt
```

Inside Vim:

1. Press **i** to enter Insert mode and type 3 lines.
2. Press **Esc** to return to Command mode.
3. Type **:set number** and press Enter.
4. Type **/TechNova** and press Enter, then **n** for the next match.
5. Press **gg** to go to line 1, then **yy** to copy the line.
6. Press **G** to go to the last line, then **p** to paste.
7. Press **u** to undo the paste.
8. Type **:wq** and press Enter.

Task 5: Users and Groups

```
sudo groupadd dev_team
sudo useradd -m -s /bin/bash alice
sudo useradd -m -s /bin/bash bob
sudo passwd alice
sudo passwd bob
sudo usermod -aG dev_team alice
sudo usermod -aG dev_team bob
groups alice bob
sudo passwd -l bob
sudo chage -l alice
sudo grep bob /etc/shadow      # a "!" before the hash means locked
```

Task 6: Permissions

```
cd ~/technova
touch scripts/backup.sh docs/secret.txt
chmod 700 scripts/backup.sh
chmod 600 docs/secret.txt

sudo mkdir /shared
sudo chgrp dev_team /shared
sudo chmod 770 /shared

ls -l scripts docs
ls -ld /shared
```

Expected output (roughly):

```
-rwx----- 1 user user 0 ... backup.sh
-rw----- 1 user user 0 ... secret.txt
drwxrwx--- 2 root dev_team 4096 ... /shared
```

Task 7: Links

```
cd ~/technova
echo "Hello Linux" > original.txt
ln original.txt hard.txt
ln -s original.txt soft.txt
ls -li
rm original.txt
cat hard.txt      # works
cat soft.txt      # error: broken link
```

Task 8: sudo

```
ls /root          # Permission denied
sudo ls /root     # works
sudo usermod -aG sudo alice
groups alice
```

Task 9: Archiving

```
cd ~
tar -czvf technova/backups/technova_backup.tar.gz technova/docs technova/scripts
tar -tvf technova/backups/technova_backup.tar.gz
mkdir -p /tmp/restore
tar -xzvf technova/backups/technova_backup.tar.gz -C /tmp/restore
ls -R /tmp/restore
```

Task 10: Cron

```
sudo systemctl status cron      # start with: sudo systemctl start cron
crontab -e
```

Add these lines (replace student with your username):

```
* * * * * /bin/touch /home/student/technova/backups/min_$(date +%Y%m%d_%H%M%S).txt
0 2 * * * tar -czf /home/student/technova/backups/docs_$(date +%F).tar.gz /home/student/technova/docs
```

Then verify:

```
crontab -l
sleep 120
ls ~/technova/backups
```

Remove the every-minute job by running `crontab -e` and deleting that line. The `\%` is required because `%` has a special meaning in `crontab`.

Task 11: Cleanup

```
sudo userdel -r alice
sudo userdel -r bob
sudo groupdel dev_team
sudo rmdir /shared
rm -rf /tmp/restore
```

PART 3: QUESTIONS AND ANSWERS

Q1. What does \$ vs # at the prompt mean?

\$ means a normal user and # means the root user.

Q2. What is the difference between cp and mv?

cp makes a copy and keeps the original. mv moves or renames the file, so the original location no longer has it.

Q3. Why did we use mkdir -p?

It creates nested directories in one command and does not give an error if a directory already exists.

Q4. Which Vim mode did you use to type text, and how do you exit it?

Insert mode (press i). Exit with Esc.

Q5. How do you save and quit Vim? How do you quit without saving?

Save and quit with :wq. Quit without saving with :q!

Q6. What is the difference between useradd and usermod -aG?

useradd creates a new user. usermod -aG adds an existing user to extra groups. The -a means append, so existing groups are not removed.

Q7. Which files store user and group information?

/etc/passwd for user accounts, /etc/shadow for encrypted passwords, /etc/group for groups, and /etc/gshadow for secure group data.

Q8. What does chmod 750 mean?

Owner has rwx (7), group has r-x (5), and others have no access (0).

Q9. What does -rw-r--r-- mean?

It is a regular file. The owner can read and write, the group can only read, and others can only read.

Q10. Why did soft.txt break but hard.txt still work after deleting original.txt?

A hard link points to the same inode, so the data stays until all links are removed. A soft link points to the file name, so it breaks when the name is deleted.

Q11. What is the difference between a hard link and a soft link?

A hard link shares the inode, cannot cross file systems, and cannot link directories. A soft link is a pointer to a path, can cross file systems, and can link directories.

Q12. Why use sudo instead of logging in as root?

It limits privileges, prevents accidental damage, and keeps an audit trail of who did what.

Q13. What is the difference between tar -czvf and tar -xzvf?

-c creates a gzip-compressed archive. -x extracts it. -z means gzip, -v is verbose, and -f gives the file name.

Q14. Which is the fastest compression: gzip, bzip2, or xz? Which compresses best?

gzip is the fastest. xz compresses best but is the slowest.

Q15. What does 0 2 * * * mean in cron?

It runs every day at 2:00 AM.

Q16. What are the five cron fields in order?

Minute, hour, day of month, month, day of week.

Q17. Why must % be written as \% in crontab?

Cron treats an unescaped % as a newline, so it must be escaped inside commands.

Q18. What does passwd -l username do?

It locks the account by putting ! before the encrypted password in /etc/shadow.

Q19. What is the link count of a new empty directory, and why?

It is 2: one for the directory's own name and one for its . entry.

Q20. Why is rm -rf dangerous?

It deletes files and directories forcefully with no confirmation and no recovery, so one wrong path can destroy important data.

Grading Rubric (100 marks)

Section	Marks
Tasks 1-3 (basics, files)	20
Task 4 (Vim)	10
Tasks 5-6 (users, permissions)	25
Tasks 7-8 (links, sudo)	15
Tasks 9-10 (tar, cron)	20
Task 11 + Q&A	10